

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *prev;
7      struct Node *next;
8  };
9
10 struct Node* createNode(int data) {
11     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
12     newNode->data = data;
13     newNode->prev = NULL;
14     newNode->next = NULL;
15     return newNode;
16 }
17
18 void createList(struct Node **head, int data) {
19     struct Node *newNode = createNode(data);
20
21     if (*head == NULL) {
22         *head = newNode;
23         return;
24     }
25     struct Node *temp = *head;
26     while (temp->next != NULL)
27         temp = temp->next;
28
29     temp->next = newNode;
30     newNode->prev = temp;
31 }
32
33 void insertLeft(struct Node **head, int value, int newValue) {
34     struct Node *temp = *head;
35
36     while (temp != NULL && temp->data != value)
37         temp = temp->next;
38
39     if (temp == NULL) {
40         printf("Node with value %d not found.\n", value);
41         return;
42     }
43
44     struct Node *newNode = createNode(newValue);
45
46     newNode->next = temp;
47     newNode->prev = temp->prev;
48
49     if (temp->prev != NULL)
50         temp->prev->next = newNode;
51     else
```

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```
49     if (temp->prev != NULL)
50         temp->prev->next = newNode;
51     else
52         *head = newNode;
53
54     temp->prev = newNode;
55
56     printf("Inserted %d to the left of %d\n", newValue, value);
57 }
58
59 void deleteValue(struct Node **head, int value) {
60     struct Node *temp = *head;
61
62     while (temp != NULL && temp->data != value)
63         temp = temp->next;
64
65     if (temp == NULL) {
66         printf("Value %d not found.\n", value);
67         return;
68     }
69
70     if (temp->prev != NULL)
71         temp->prev->next = temp->next;
72     else
73         *head = temp->next;
74
75     if (temp->next != NULL)
76         temp->next->prev = temp->prev;
77
78     free(temp);
79     printf("Node with value %d deleted.\n", value);
80 }
81
82 void display(struct Node *head) {
83     if (head == NULL) {
84         printf("List is empty.\n");
85         return;
86     }
87     printf("Doubly Linked List: ");
88     struct Node *temp = head;
89
90     while (temp != NULL) {
91         printf("%d <-> ", temp->data);
92         temp = temp->next;
93     }
94     printf("NULL\n");
95 }
96
97 int main() {
98     struct Node *head = NULL;
99     int choice, value, newValue;
```

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```
90 while (temp != NULL) {
91     printf("%d <-> ", temp->data);
92     temp = temp->next;
93 }
94 printf("NULL\n");
95 }
96
97 int main() {
98     struct Node *head = NULL;
99     int choice, value, newValue;
100
101     while (1) {
102         printf("\n--- MENU ---\n");
103         printf("1. Create\n");
104         printf("2. Insert left of node\n");
105         printf("3. Delete node by value\n");
106         printf("4. Display list\n");
107         printf("5. Exit\n");
108         printf("Enter your choice: ");
109         scanf("%d", &choice);
110
111         switch (choice) {
112             case 1:
113                 printf("Enter value to insert: ");
114                 scanf("%d", &value);
115                 createList(&head, value);
116                 break;
117             case 2:
118                 printf("Enter existing node value: ");
119                 scanf("%d", &value);
120                 printf("Enter new value to insert left: ");
121                 scanf("%d", &newValue);
122                 insertLeft(&head, value, newValue);
123                 break;
124             case 3:
125                 printf("Enter value to delete: ");
126                 scanf("%d", &value);
127                 deleteValue(&head, value);
128                 break;
129             case 4:
130                 display(head);
131                 break;
132             case 5:
133                 exit(0);
134             default:
135                 printf("Invalid choice!\n");
136         }
137     }
138     return 0;
139 }
140
```

—

"C:\Users\Admin\Desktop\18M24CS173\Double Linked list.exe"

```
5. Exit
Enter your choice: 1
Enter value to insert: 18

--- MENU ---
1. Create
2. Insert left of node
3. Delete node by value
4. Display list
5. Exit
Enter your choice: 4
Doubly Linked List: 45 <-> 18 <-> NULL

--- MENU ---
1. Create
2. Insert left of node
3. Delete node by value
4. Display list
5. Exit
Enter your choice: 2
Enter existing node value: 18
Enter new value to insert left: 17
Inserted 17 to the left of 18

--- MENU ---
1. Create
2. Insert left of node
3. Delete node by value
4. Display list
5. Exit
Enter your choice: 4
Doubly Linked List: 45 <-> 17 <-> 18 <-> NULL

--- MENU ---
1. Create
2. Insert left of node
3. Delete node by value
4. Display list
5. Exit
Enter your choice: 3
Enter value to delete: 45
Node with value 45 deleted.

--- MENU ---
1. Create
2. Insert left of node
3. Delete node by value
4. Display list
5. Exit
Enter your choice: 4
Doubly Linked List: 17 <-> 18 <-> NULL

--- MENU ---
1. Create
2. Insert left of node
3. Delete node by value
4. Display list
5. Exit
Enter your choice: 5

Process returned 0 (0x0)   execution time : 63.442 s
Press any key to continue.
```

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